

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization

Lecture No.- 2

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Recap of Previous Lecture



Topic

Peripheral Device

Topic

IO vs Memory Buses

Topic

Memory Mapped IO vs IO Mapped IO

Topic

Asynchronous Data Transfer

Topics to be Covered



Topic

Modes of Transfer

Topic

Programmed IO

Topic

Interrupt IO



Topic : Modes of Transfer

1. Programmed I/O or Program Controlled I/O
2. Interrupt Initiated I/O or Interrupt Driven I/O
3. Direct memory Access (DMA)



Topic : Programmed IO

- There is no any provision through which IO can inform to CPU about data transfer
- IO sets its own status and waits
- CPU runs program periodically and checks the status of each device one-by-one ^{→ polling}
- If any device has its status set then CPU performs data transfer for it.

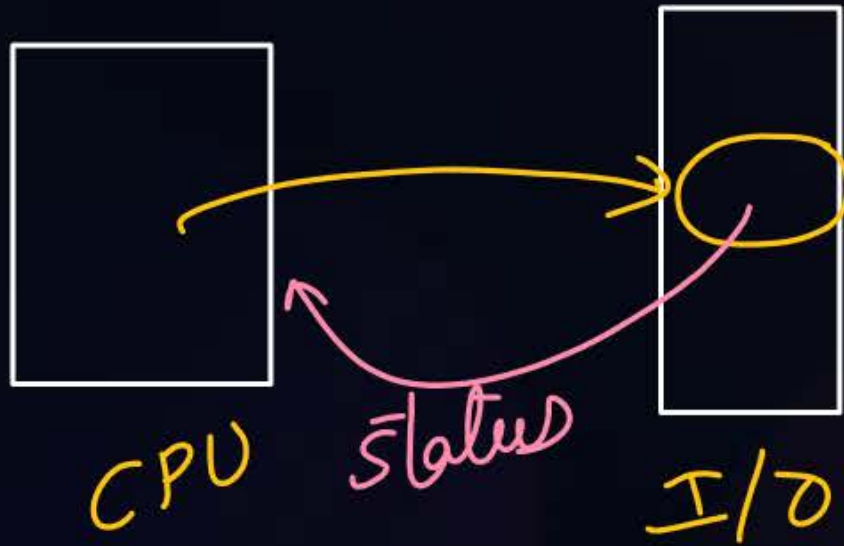
☹️ ⇒ wastage of CPU time



Topic : Time Required in Programmed IO

$$= \text{status check time} + \text{data transfer time}$$

depends on I/O's speed





Topic : Interrupt Initiated IO

- IO device has a provision (Interrupt Signal) to inform to CPU about communication.



Topic : Interrupt Initiated IO

- IO device has a provision (Interrupt Signal) to inform to CPU about communication.
- When CPU receives interrupt:
 - It completes execution of current instruction
 - Saves the status (PC, PSW etc.) of current process onto the stack *in memory*
 - Branches to service the interrupt
 - Resumes the previous process by taking out the values from stack

#Q. The following are some events that occur after a device controller issues an interrupt while process L is under execution.

P. The processor pushes the process status of L onto the control stack

Q. The processor finishes the execution of the current instruction

R. The processor executes the interrupt service routine

S. The processor pops the process status of L from the control stack

T. The processor loads the new PC value based on the interrupt

Which of the following is the correct order in which the events above occur?

A ✓ QPTRS

QPTRS

B PTRSQ

C TRPQS

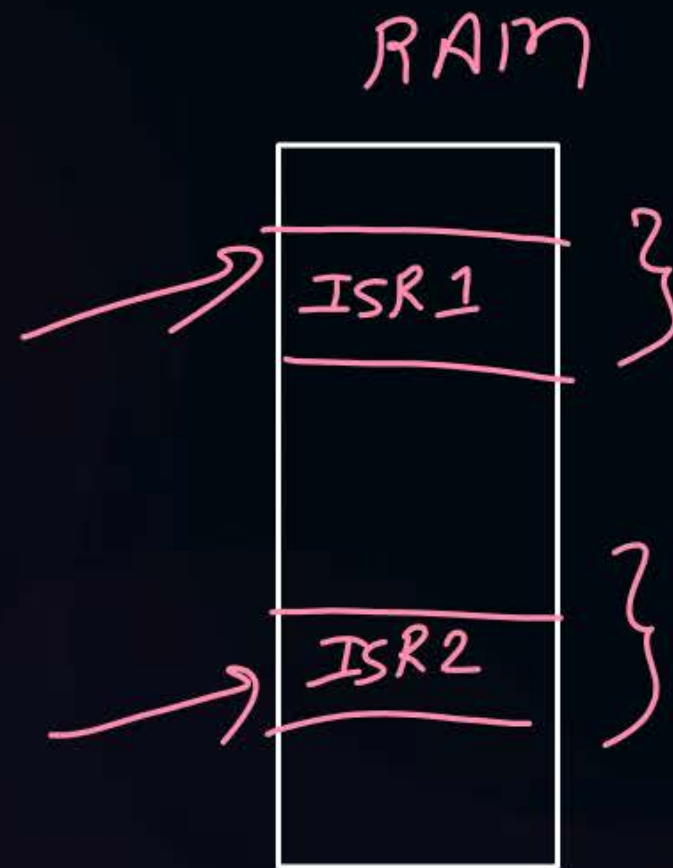
D QTPRS



Topic : Interrupt Initiated IO

ISR (Interrupt Service Routine):-

A small program segment, execution of which services the interrupt.





Topic : Vectored vs Non-Vectored

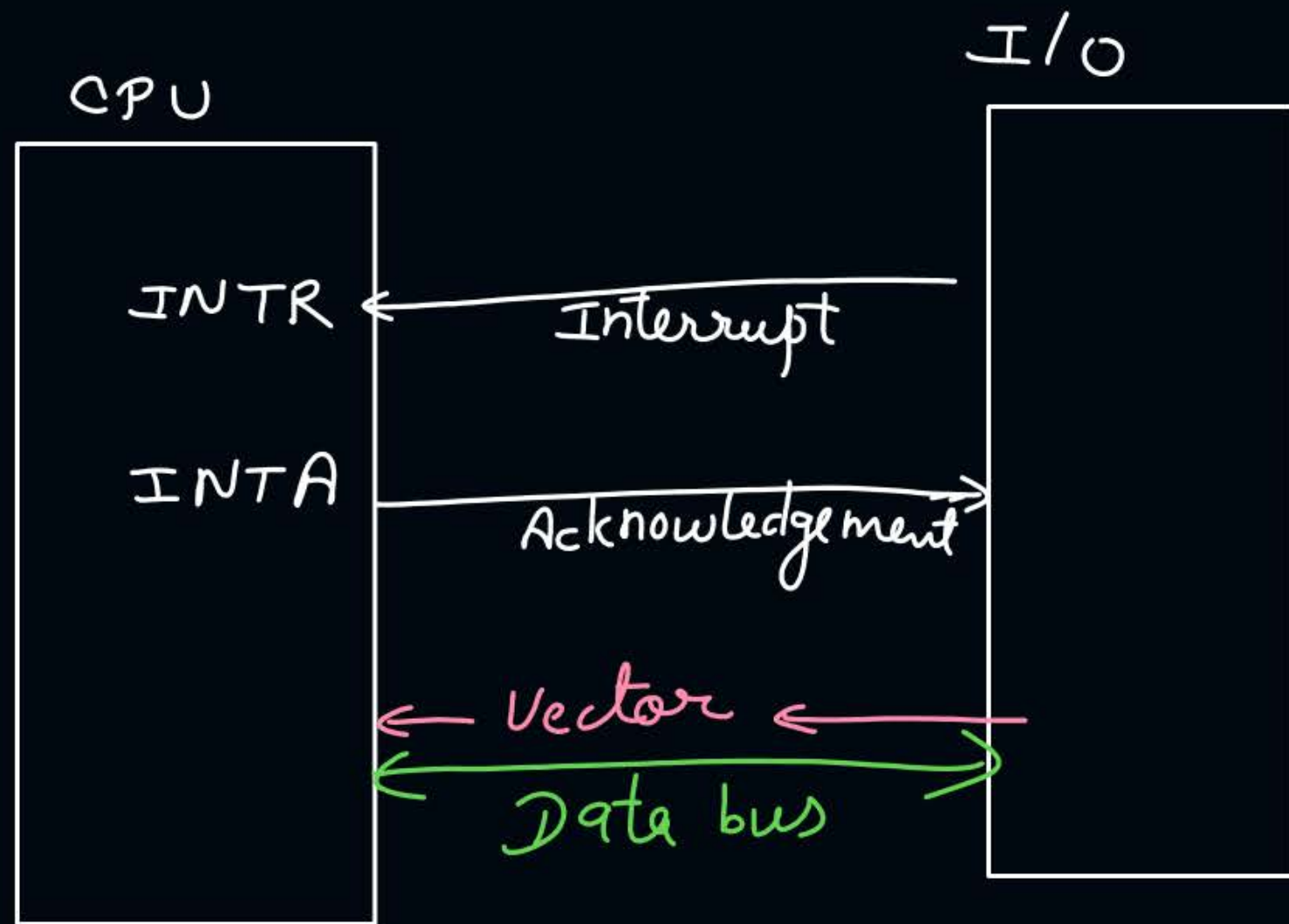
(scalar)

device sends interrupt
and reference of ISR
vector

CPU can execute ISR
& can service the interrupt

device sends only interrupt

↓
CPU runs a default service routine
first and figures out address of
actual ISR.
Then CPU runs actual ISR.





Topic : Maskable vs Non-Maskable



↓
CPU can accept or reject

↓
always accept





Topic : Internal Vs External

S/W



CPU interrupts itself
because of some unexpected
error during instruction
execution.

example:- page fault

H/W interrupt



from I/O devices



Topic : Simultaneous Interrupts

when CPU receives interrupts from multiple devices simultaneously then CPU services the interrupt as per the priority of devices.

Priority based interrupt handling solution

S/w solⁿ

CPU runs a program to figure out which device's interrupt will be serviced first.

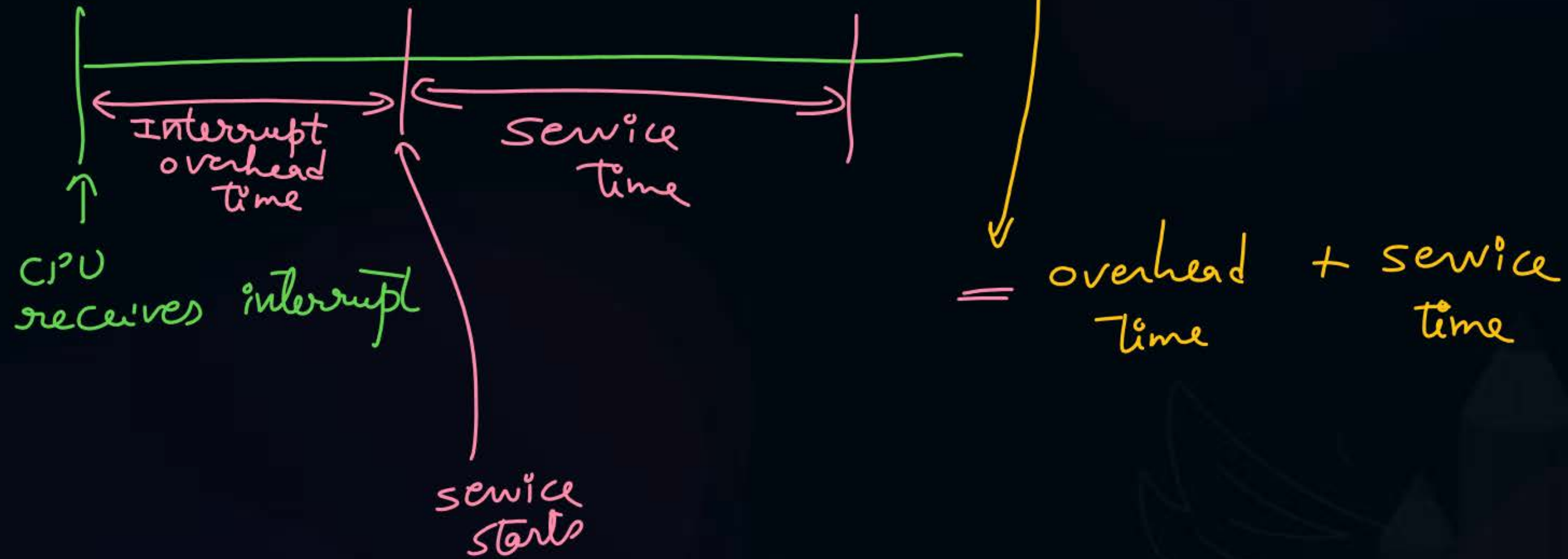
H/w solⁿ

serial
(daisy chaining)

parallel



Topic : Time Required in Interrupt IO



#Q. Consider a CPU which takes 0.05 microseconds as interrupt overhead time when a device generates interrupt for CPU, and CPU accepts it. After that CPU takes 4 cycles to service the interrupt. If CPU runs on 10MHz clock rate then total time CPU spends for interrupt service is 0.45 microseconds?

$$\begin{aligned} &= 0.05 \mu\text{sec} + 4 \text{ cycles} \\ &= 0.05 \mu\text{sec} + 4 * \frac{1}{10\text{MHz}} \\ &= 0.05 \mu\text{sec} + 0.4 \mu\text{sec} \\ &= 0.45 \mu\text{sec} \end{aligned}$$

#Q. A device with data transfer rate 20 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 10 microsecond.

1. Total time required in programmed IO for 10 bytes data transfer? $\Rightarrow 550 \mu\text{sec}$
2. Total time required in interrupt IO for 10 bytes data transfer? $\Rightarrow 510 \mu\text{sec}$
3. What is the minimum performance gain of operating the device under interrupt mode over operating it under program controlled mode? $\Rightarrow 1.0784$

for 20KB, I/O time = 1 sec

for 1B, $\frac{1 \text{ sec}}{20 \text{ KB}} = \frac{1 \text{ sec}}{20 \times 10^3 \text{ B}}$

$$= \frac{1}{20} \text{ msec} = \frac{1000}{20} \mu\text{sec} = 50 \mu\text{sec}$$

$$\begin{aligned}
 1. \quad \text{Programmed I/O time} &= \text{status check time} + \text{data transfer time} \\
 &= 50 \mu\text{sec} + (10 * 50 \mu\text{sec}) \\
 &= 550 \mu\text{sec}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \text{Interrupt I/O time} &= 10 \mu\text{sec} + (10 * 50 \mu\text{sec}) \\
 &= 510 \mu\text{sec}
 \end{aligned}$$

$$\begin{aligned}
 3. \quad \text{Performance gain} &= \frac{\text{older technique time}}{\text{newer technique time}} = \frac{550 \mu\text{sec}}{510 \mu\text{sec}} \\
 \text{or speed up of new technique} &= 1.0784
 \end{aligned}$$



Topic : DMA

(Direct mem. Access)



- Enables data transfer between I/O and memory without CPU intervention
- Need a hardware: DMAC (DMA Controller)





2 mins Summary



Topic

Modes of Transfer

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Programmed IO

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Interrupt IO



Happy Learning

THANK - YOU

